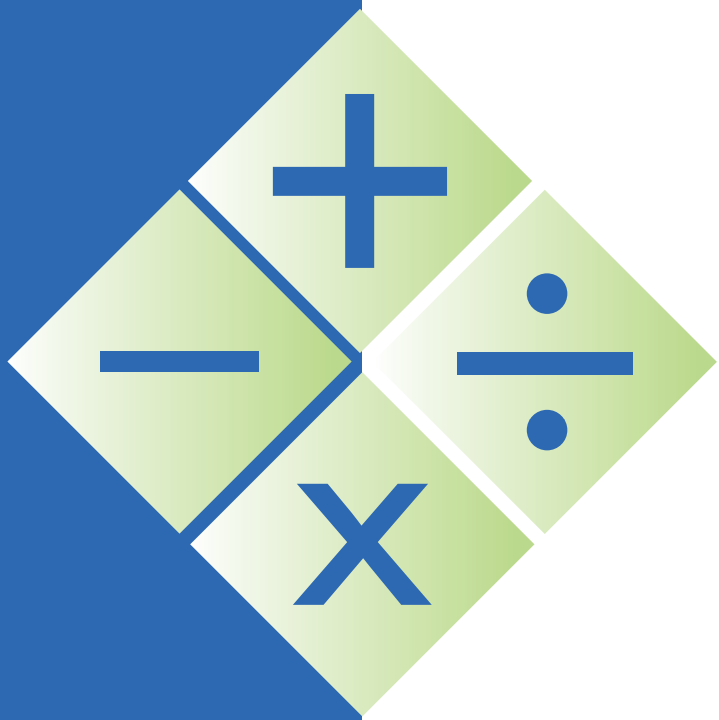
Mathematics

Forward Exam Practice Test Grade 4



Wisconsin Forward Exam

WISCONSIN DEPARTMENT OF Public Instruction

MATHEMATICS ITEMS—SESSION 1

1. Which statement is correct?

A. The comparison 0.38 < 0.71 is true because the 8 in 0.38 is greater than the 1 in 0.71.

B. The comparison 0.38 > 0.71 is true because the 8 in 0.38 is greater than the 1 in 0.71.

C. The comparison 0.38 < 0.71 is true because 0.38 is less than $\frac{1}{2}$ and 0.71 is greater than $\frac{1}{2}$.

D. The comparison 0.38 > 0.71 is true because 0.38 is less than $\frac{1}{2}$ and 0.71 is greater than $\frac{1}{2}$.

2. A basketball team played two games.

- The number of tickets sold for the first game was 29,294.
- The number of tickets sold for the second game was 29,212.

Select the **two** statements that describe the **most reasonable** estimates of the total number of tickets sold for both games.

Select **two** options.

A. An estimate of 48,000 is a little less than the actual number of tickets because both numbers are slightly more than 20,000.

B. An estimate of 58,000 is a little less than the actual number of tickets because both numbers are rounded down to arrive at the estimate.

C. An estimate of 60,000 is a little less than the actual number of tickets because both numbers are rounded down to arrive at the estimate.

D. An estimate of 60,000 is a little greater than the actual number of tickets because both numbers are slightly less than 30,000.

E. An estimate of 80,000 is a little greater than the actual number of tickets because both numbers are rounded up to arrive at the estimate.

Go on to the next page.

5. Match each decimal number to an equivalent fraction.

0.79	$\frac{97}{100}$
0.7	$9\frac{7}{10}$
0.97	$\frac{7}{10}$
9.7	$\frac{79}{100}$

6. Shirley is 8 years old. Shirley's uncle is 6 times as old as Shirley. Write an expression that could be used to represent how many years old Shirley's uncle is.

Go on to the next page.

7. At a sandwich shop, Lionel orders a large sandwich and Marissa orders a small sandwich.

- Lionel saves $\frac{1}{2}$ of his sandwich to eat later.
- Marissa saves $\frac{1}{4}$ of her sandwich to eat later.

Lionel says that together they saved $\frac{3}{4}$ of a whole sandwich. Which statement **best** explains whether Lionel is correct?

A. Lionel is correct because $\frac{1}{2} = \frac{2}{4}$, and $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$.

B. Lionel is correct because he has 1 piece of sandwich left and Marissa has 3 pieces left, and 1 + 3 = 4.

C. Lionel is not correct because $\frac{1}{2} + \frac{1}{4}$ is not equal to $\frac{3}{4}$.

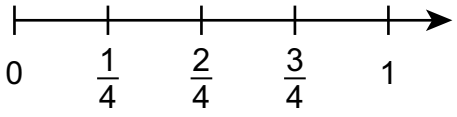
D. Lionel is not correct because the two sandwiches were different sizes to begin with.

8. Henry is growing tomatoes. The weight, in pounds, of each tomato is listed.

$$\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}$$

Create a line plot to display Henry's data.

Tomatoes



Weight (pounds)

Go on to the next page.

9. A museum gift shop has postcards printed. There are 18 different designs for the postcards. The gift shop has 84 postcards of each design printed. How many postcards, in total, does the gift shop have printed?

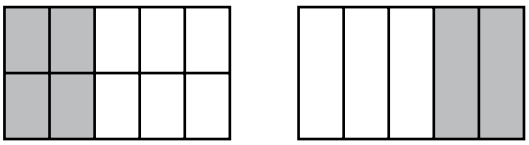
A. 102

B. 216

C. 882

D. 1,512

10. Fraction model A and fraction model B are each shaded to represent a fraction as shown. The models are the same size.



fraction model A fraction model B

Which statement about the fraction models is correct?

A. The fraction $\frac{4}{10}$ is greater than the fraction $\frac{2}{5}$ because 4 is greater than 2 and 10 is greater than 5.

B. Fraction model A has more shaded parts than fraction model B, so the fractions $\frac{4}{10}$ and $\frac{2}{5}$ are not equivalent.

C. The fraction $\frac{4}{10}$ is less than the fraction $\frac{2}{5}$ because the shaded parts of fraction model B are larger than the shaded parts of fraction model A.

D. The total amount shaded in fraction model A is the same as the total amount shaded in fraction model B, so the fractions $\frac{4}{10}$ and $\frac{2}{5}$ are equivalent.

Go on to the next page.

11. Julie creates a number pattern that starts with 3 and uses the rule "add 7." Which statement about Julie's number pattern is correct?

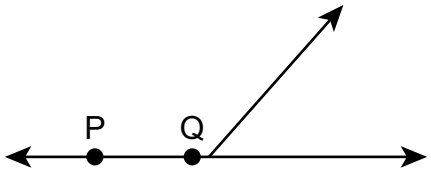
A. All the numbers in the pattern will be odd.

B. All the numbers in the pattern will be even.

C. The numbers in the pattern will alternate between odd and even.

D. The first number in the pattern will be odd but the rest will be even.

12. The figure shows the location of two points, P and Q.



Which term **best** describes the part of the figure located between points P and Q?

A. line

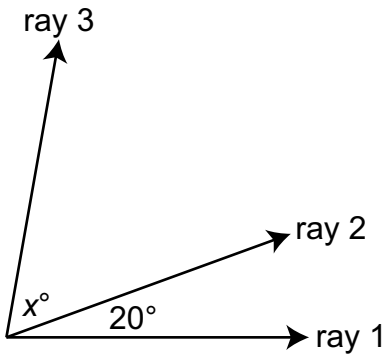
B. line segment

C. point

D. ray

Go on to the next page.

13. An angle is shown.



The measure of the angle formed by ray 1 and ray 3 is 80°. Use the numbers, variables, and symbols below the blank lines to make an equation that could be used to find the value of *x*.

_____	_____	_____	=	_____
0	+	0		0
20	-	20		20
80	×	80		80
x	÷	*x*		*x*

Go on to the next page.

14. Which expression is equivalent to $\frac{5}{6}$?

A. $5 + \frac{1}{6}$

B. $5 \times \frac{1}{6}$

C. $6 \times \frac{1}{5}$

D. $\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$

15. Several students are playing a video game.

- Yuri scores seven hundred forty-five points.
- Hudson scores seven hundred ninety-one points.
- Amelia scores more points than Yuri but less points than Hudson.

What is a possible number of points that Amelia could have scored?

STOP.

MATHEMATICS ITEMS—SESSION 2

1. For each number in the table, determine whether it is prime or composite.

	Prime	Composite
2		
7		
15		
27		

2. An apartment complex has 6 buildings, and the number of people who live in each building is shown in the table.

Building	Number of People
A	1,152
B	894
C	491
D	813
E	706
F	1,015

Which number is the **best** estimate for the number of people who live in the entire apartment complex?

A. 3,000

B. 5,000

C. 6,000

D. 30,000

Go on to the next page.

3. Several students are running a race.

- Jolie finished $\frac{43}{100}$ of a second faster than Frank.
- Frank finished $\frac{2}{10}$ of a second faster than Ingrid.

What fraction of a second faster than Ingrid was Jolie?



4. Demarcus worked on his science project for 3 hours. For how many **minutes** did he work on his project?

A. 60

B. 63

C. 120

D. 180

5. Select the **two** statements that represent $54 = 6 \times 9$.

Select **two** options.

A. The number 54 is 6 less than the number 9.

B. The number 54 is 6 more than the number 9.

C. The number 54 is 9 more than the number 6.

D. The number 54 is 6 times as many as the number 9.

E. The number 54 is 9 times as many as the number 6.

Go on to the next page.

6. A recipe calls for $1\frac{3}{4}$ cups of milk. Which expression represents one way to measure the milk for the recipe and is equivalent to $1\frac{3}{4}$?

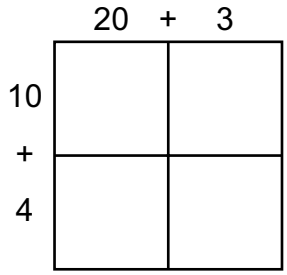
A. $\frac{1}{4}+\frac{1}{4}+\frac{1}{4}$

B. $\frac{1}{4}+\frac{1}{4}+\frac{1}{4}+\frac{1}{4}$

C. $\frac{4}{4}+\frac{1}{4}+\frac{1}{4}+\frac{1}{4}$

D. $\frac{4}{4}+\frac{1}{4}+\frac{1}{4}+\frac{1}{4}+\frac{1}{4}$

7. Write a number from the list in each box in the area model to show 14 × 23.

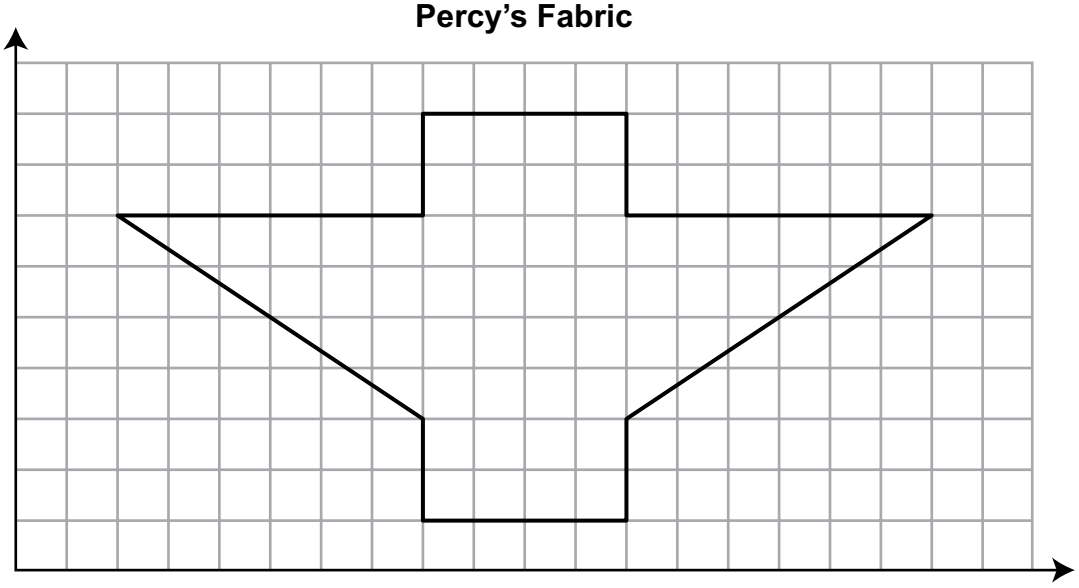


3 10 12 14 20 23 30 80 200

Go on to the next page.

8. Percy cuts out a piece of fabric in the shape shown on the grid. He folds the fabric in half so both halves match. Draw a line across the shape to show where Percy folds the fabric.

Percy's Fabric



9. Amruta has 2 pencils on her desk. She has 4 times as many pencils in her pencil case as on her desk. Which statement describes one way to find the number of pencils in Amruta's pencil case?

A. Subtract 2 from 4 because 2 out of 4 pencils are on the desk.

B. Multiply 2 and 4 because there are 4 times as many as 2 pencils.

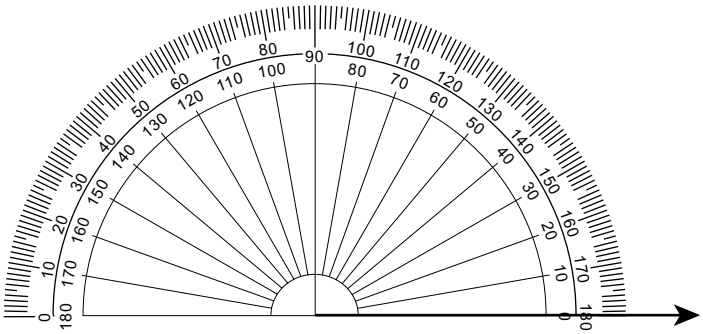
C. Divide 4 by 2 because the pencils are split into two different groups.

D. Add 4 and 2 because there is a set of 2 pencils and a set of 4 pencils.

Go on to the next page.

10. Derek has had a pet lizard for 1,163 days. He has had his pet fish for 675 fewer days than he has had his pet lizard. How many days has Derek had his pet fish?

11. Using the ray and the protractor shown, draw a 135° angle.



Go on to the next page.

12. Some students are painting a mural at their school. Each student has a section to paint, and each section is the same size. The table shows the fraction of each section several students have completed.

Student	Fraction Completed
Aaron	$\frac{4}{5}$
Bonnie	$\frac{2}{5}$
Cam	$\frac{4}{8}$
Drea	$\frac{1}{8}$

Which student has completed the most of their section of the mural?

A. Aaron

B. Bonnie

C. Cam

D. Drea

13. Bianca correctly solved 9 × 7 by decomposing it into two expressions. Write a number from the list in each box to complete the expression to show how Bianca decomposed 9 × 7.

$$(5 \times 7) + (\square \times \square)$$

1 2 3 4 5 6 7 8 9 10

Go on to the next page.

14. There are 12,239 students in the local school district. How many times greater is the value of the underlined 2 than the value of the 2 that is not underlined?

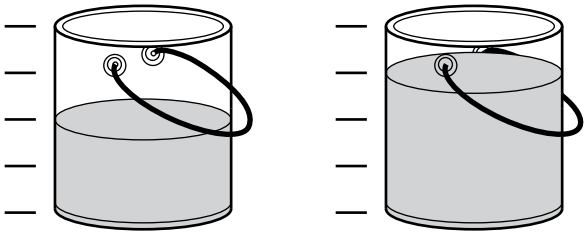
A. 10

B. 20

C. 100

D. 1,000

15. Jalen has two containers of paint. Each container can hold 1 gallon of paint when full. The picture shows the amount of paint, in gallons, inside each of Jalen's containers.



What fraction represents the total amount of paint, in gallons, Jalen has inside these two containers?

STOP.

MATHEMATICS—APPENDIX

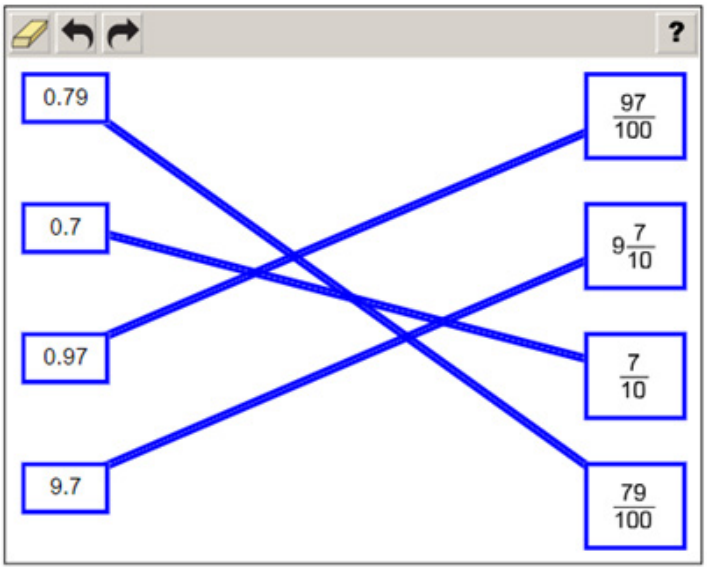
SUMMARY DATA—GRADE 4, SESSION 1

Number 1	
Alignment	M.4.NF.C.7
Depth of Knowledge	2
Key(s)	C
Points	1
Annotations	The question asks the student to determine a correct comparison of two decimals and a correct explanation. A. Incorrect. The student chooses the correct comparison, but the explanation compares the hundredths places. B. Incorrect. The student chooses the incorrect comparison and compares only the hundredths places. C. Correct. The decimals can be compared by using a comparison to the unit fraction $\frac{1}{2}$. D. Incorrect. The student chooses the incorrect comparison but the correct explanation.

Number 2	
Alignment	M.4.NBT.A.3
Depth of Knowledge	2
Key(s)	B, D
Points	2
Annotations	The question asks the student to determine a reasonable estimate of a sum of two numbers. To receive full credit, the student must select choices B and D. To receive partial credit, the student must select either choice B or choice D. A. Incorrect. The student did not use appropriate estimates for the given numbers, resulting in an estimate of the sum that is too low. B. Correct. Both values are slightly greater than 29,000, and 29,000 + 29,000 = 58,000, so 58,000 is a little less than the actual sum. C. Incorrect. The student used an appropriate estimated sum, but the reasoning about the estimate of the sum is incorrect. D. Correct. Both values are slightly less than 30,000, so the sum would be slightly less than 60,000. E. Incorrect. The student used an estimate too high for the given numbers.

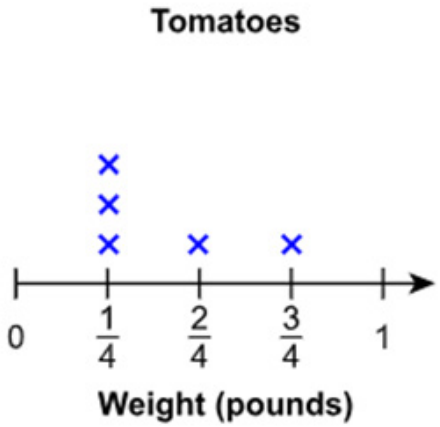
Number 3	
Alignment	M.4.OA.A.3
Depth of Knowledge	2
Key(s)	93
Points	1
Annotations	The question asks the student to solve a multistep word problem. To receive full credit, the student must enter 93 or an equivalent value.

Number 4	
Alignment	M.4.G.A.3
Depth of Knowledge	1
Key(s)	B
Points	1
Annotations	The question asks the student to identify a shape without a line of symmetry. A. Incorrect. The student chooses a shape with more than one line of symmetry. B. Correct. The shape does not have any lines of symmetry. C. Incorrect. The student chooses a shape with one line of symmetry. D. Incorrect. The student chooses a shape with more than one line of symmetry.

Number 5	
Alignment	M.4.NF.C.6
Depth of Knowledge	1
Key(s)	See Annotations
Points	1
Annotations	The question asks the student to match a decimal with an equivalent fraction. To receive full credit, the student must match each decimal number to the equivalent fraction, as shown. 

Number 6	
Alignment	M.4.OA.A.1
Depth of Knowledge	1
Key(s)	8 × 6 OR 6 × 8
Points	1
Annotations	The question asks the student to create an expression to represent a multiplicative comparison. To receive full credit, the student must enter the expression 8 × 6 OR 6 × 8.

Number 7	
Alignment	M.4.NF.B.3a
Depth of Knowledge	2
Key(s)	D
Points	1
Annotations	The question asks the student to determine the correctness of a statement by reasoning about a given sum of fractions. A. Incorrect. The student adds the fractions without regard to the sizes of the wholes. B. Incorrect. The student adds the number of parts of the sandwich without regard to the sizes of the wholes. C. Incorrect. The student incorrectly adds the fractions. D. Correct. The fractions can not be added because they are representative of different sized wholes.

Number 8	
Alignment	M.4.MD.B.4
Depth of Knowledge	1
Key(s)	See Annotations
Points	1
Annotations	The question asks the student to create a line plot. To receive full credit, the student must create a line plot with the 5 given fractions. 

Number 9	
Alignment	M.4.NBT.B.5
Depth of Knowledge	2
Key(s)	D
Points	1
Annotations	The question asks the student to find the total number of items. A. Incorrect. The student adds the two values. B. Incorrect. The student multiplies 18 × 8 and 18 × 4 then adds the two values. C. Incorrect. The student incorrectly multiplies 84 × 8. D. Correct. The total is found by multiplying the two given numbers.

Number 10	
Alignment	M.4.NF.A.1a
Depth of Knowledge	2
Key(s)	D
Points	1
Annotations	The question asks the student to compare the fractions represented in two same-sized fraction models. A. Incorrect. The student chooses the statement that compares the values of the digits. B. Incorrect. The student chooses the statement that compares the numbers of parts shaded. C. Incorrect. The student chooses the statement that compares the size of each part. D. Correct. The shaded parts are equivalent because the total amount shaded in each model is the same.

Number 11	
Alignment	M.4.OA.C.5
Depth of Knowledge	2
Key(s)	C
Points	1
Annotations	The question asks the student to use a rule for a number pattern. A. Incorrect. The student recognizes the starting value and rule are odd numbers and assumes all the numbers in the pattern will be odd. B. Incorrect. The student uses the two odd numbers to create an even number and applies the observation to the entire pattern. C. Correct. The student observes that the pattern will start with 3, 10, 17, and 24. D. Incorrect. The student uses the second number in the pattern as the observation for the rest of the pattern.

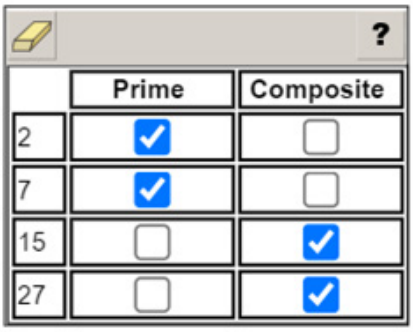
Number 12	
Alignment	M.4.G.A.1
Depth of Knowledge	1
Key(s)	B
Points	1
Annotations	The question asks the student to describe the part of a figure located between two points. A. Incorrect. The student describes the part of the figure that includes more than just the part between points P and Q. B. Correct. A line segment is a line between two points. C. Incorrect. The student describes what P and Q represent. D. Incorrect. The student describes the part of the figure that passes through point P or starts at P and passes through point Q.

Number 13	
Alignment	M.4.MD.C.7
Depth of Knowledge	2
Key(s)	See Annotations
Points	1
Annotations	The question asks the student to create an equation from the values given in an angle. To receive full credit, the student must enter the equation $x + 20 = 80$ OR $20 + x = 80$ OR $80 - 20 = x$ OR $80 - x = 20$.

Number 14	
Alignment	M.4.NF.B.4a
Depth of Knowledge	1
Key(s)	B
Points	1
Annotations	The question asks the student to determine an expression equivalent to a given fraction. A. Incorrect. The student uses addition instead of multiplication. B. Correct. The fraction $\frac{5}{6}$ can be rewritten as the product of the whole number 5 and the unit fraction $\frac{1}{6}$. C. Incorrect. The student reverses the whole number and the denominator of the fraction. D. Incorrect. The student adds only 4 of the $\frac{1}{6}$ fractions.

Number 15	
Alignment	M.4.NBT.A.2
Depth of Knowledge	1
Key(s)	See Annotations
Points	1
Annotations	The question asks the student to determine a number between two numbers. To receive full credit, the student must enter a value between 745 and 791.

SUMMARY DATA—GRADE 4, SESSION 2

Number 1	
Alignment	M.4.OA.B.4
Depth of Knowledge	1
Key(s)	See Annotations
Points	1
Annotations	The question asks the student to determine whether numbers are prime or composite. To receive full credit, the student must select that 2 is prime, 7 is prime, 15 is composite, and 27 is composite as shown. 

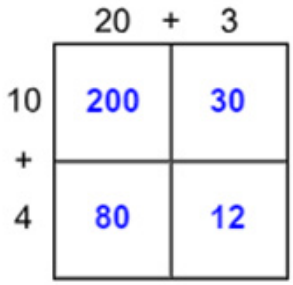
Number 2	
Alignment	M.4.NBT.A.3
Depth of Knowledge	1
Key(s)	B
Points	1
Annotations	The question asks the student to determine the best estimate for the sum of numbers in a table. A. Incorrect. The student chooses an estimate too low for the sum. B. Correct. It can be estimated that 2,000 people live in the first two buildings, 1,000 people live in the next two buildings, and 2,000 people live in the last two buildings for an estimated total of 5,000 people. C. Incorrect. The student doesn't realize that each building would need to have 1,000 people living in it to have an estimated total of 6,000. Many buildings have less than 1,000 people. D. Incorrect. The student chooses the estimate that represents multiplying each building by 6 and then finding the sum.

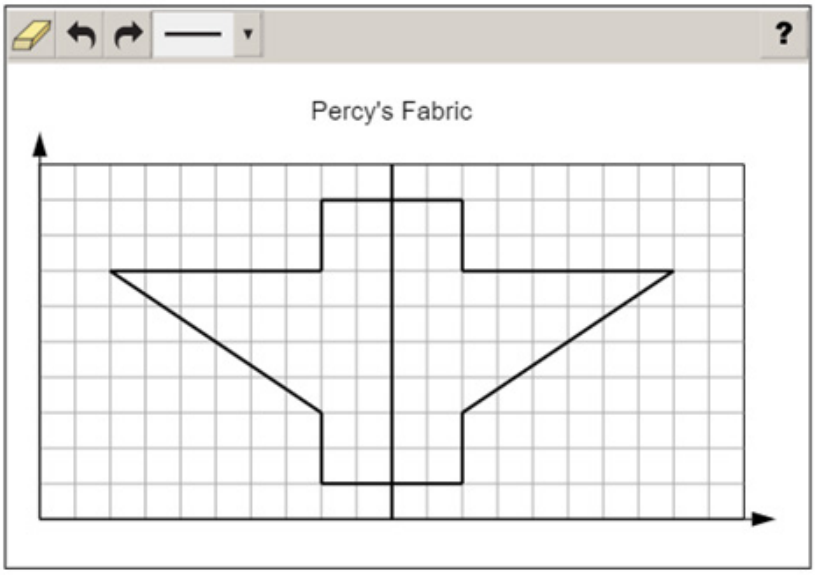
Number 3	
Alignment	M.4.NF.C.5
Depth of Knowledge	1
Key(s)	$\frac{63}{100}$
Points	1
Annotations	The question asks the student to find the sum of two fractions with denominators of 10 and 100. To receive full credit, the student must enter the value $\frac{63}{100}$ or an equivalent fraction.

Number 4	
Alignment	M.4.MD.A.1
Depth of Knowledge	2
Key(s)	D
Points	1
Annotations	The question asks the student to convert hours to minutes. A. Incorrect. The student chooses the number of minutes in 1 hour. B. Incorrect. The student chooses the number found by adding the number of minutes in 1 hour and the 3 from the 3 hours. C. Incorrect. The student chooses the number for if there were 40 minutes in 1 hour. D. Correct. The number of minutes in an hour is found by multiplying the given number of hours by 60.

Number 5	
Alignment	M.4.OA.A.1
Depth of Knowledge	1
Key(s)	D, E
Points	2
Annotations	The question asks the student to represent a multiplicative comparison in more than one way. To receive full credit, the student must select choices D and E. To receive partial credit, the student must select either choice D or choice E. A. Incorrect. The student uses subtraction instead of multiplication. B. Incorrect. The student uses addition instead of multiplication. C. Incorrect. The student uses addition instead of multiplication. D. Correct. The multiplicative equation of 54 = 6 × 9 can be represented by "The number 54 is 6 times as many as the number 9." E. Correct. The multiplicative equation of 54 = 6 × 9 can be represented by "The number 54 is 9 times as many as the number 6."

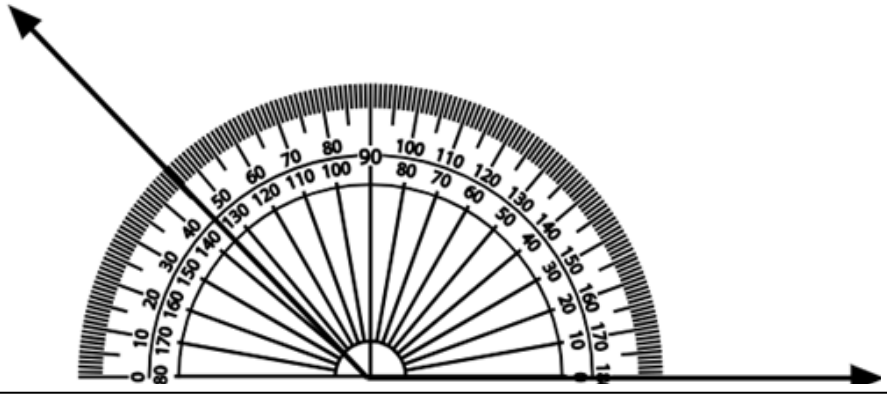
Number 6	
Alignment	M.4.NF.B.3b
Depth of Knowledge	1
Key(s)	C
Points	1
Annotations	The question asks the student to determine which addition expression represents a given fraction. A. Incorrect. The student chooses the expression that omits the whole number. B. Incorrect. The student chooses the expression that uses an incorrect fraction to represent the whole number. C. Correct. The addition expression that represents the given fraction uses $\frac{4}{4}$ to represent the whole number 1 and $\frac{1}{4}+\frac{1}{4}+\frac{1}{4}$ to represent $\frac{3}{4}$. D. Incorrect. The student chooses the expression with $\frac{1}{4}$ too many.

Number 7	
Alignment	M.4.NBT.B.5
Depth of Knowledge	2
Key(s)	See Annotations
Points	1
Annotations	The question asks the student to complete the numbers in an area model to show a multiplication problem. To receive full credit, the student must enter the four numbers in the correct boxes as shown below.  3 10 12 14 20 23 30 80 200

Number 8	
Alignment	M.4.G.A.3
Depth of Knowledge	1
Key(s)	See Annotations
Points	1
Annotations	The question asks the student to create a line of symmetry. To receive full credit, the student must draw a vertical line down the center of the figure, starting at 10 units over to the right. 

Number 9	
Alignment	M.4.OA.A.2
Depth of Knowledge	2
Key(s)	B
Points	1
Annotations	The question asks the student to solve a word problem involving a multiplicative comparison. A. Incorrect. The student uses subtraction instead of multiplication. B. Correct. The multiplicative comparison is 2 × 4 because there are 4 times as many as 2 pencils. C. Incorrect. The student uses division instead of multiplication. D. Incorrect. The student uses addition instead of multiplication.

Number 10	
Alignment	M.4.NBT.A.4
Depth of Knowledge	1
Key(s)	488
Points	1
Annotations	The question asks the student to determine the difference between two numbers. To receive full credit, the student must enter 488 or an equivalent value.

Number 11	
Alignment	M.4.MD.C.6
Depth of Knowledge	2
Key(s)	See Annotations
Points	1
Annotations	The question asks the student to draw an angle with the given measure by using the protractor tool. To receive full credit, the student must draw a 135° angle as shown. 

Number 12	
Alignment	M.4.NF.A.2
Depth of Knowledge	2
Key(s)	A
Points	1
Annotations	The question asks the student to compare fractions with like and unlike denominators. A. Correct. The student chooses the fraction with the greatest value. B. Incorrect. The student chooses a fraction that has a value close to one-half. C. Incorrect. The student chooses the fraction with the greatest denominator and compares it to the next fraction. D. Incorrect. The student chooses the fraction with the least value.

Number 13	
Alignment	M.4.OA.D.6
Depth of Knowledge	2
Key(s)	See Annotations
Points	1
Annotations	The question asks the student to decompose an expression into two expressions. To receive full credit, the student must enter 4 and 7 or 7 and 4.

Number 14	
Alignment	M.4.NBT.A.1
Depth of Knowledge	1
Key(s)	A
Points	1
Annotations	The question asks the student to determine how many times greater the value of one number is than the value of another in a given number. A. Correct. The value of the 2 in the thousands place is 10 times greater than the 2 in the hundreds place. B. Incorrect. The student multiplied the correct answer by the digit, 2. C. Incorrect. The student gave the value of the 2 that is not underlined. D. Incorrect. The student chooses the place value of the underlined 2.

Number 15	
Alignment	M.4.NF.B.3d
Depth of Knowledge	2
Key(s)	See Annotations
Points	1
Annotations	The question asks the student to add two fractions. To receive full credit, the student must enter the fraction $\frac{5}{4}$ or an equivalent value.